\

Course Code: SECI1143-03
Lecturer's Name: Dr Anis Farhan Binti Kamaruzaman

Group Name: False 9

Group members:

Name	No Matric
Muhammad Najmi Shahmi Bin Mohd Latpi	A25CS0279
Nasaaie Bin Noriskamar	A25CS0118
Akmal Rafique Bin Ahmad Raphaie	A25CS0181
Muhammad Fairuz Bin Herman	A25CS0267

Table Of Content

1. Introduction

This study case was conducted to delve into student’s engagement in physical activities and their time spent on social media. In recent years, the use of social media has statistically increase among university students which might have a negative impact on their lifestyle in terms of physical activities.

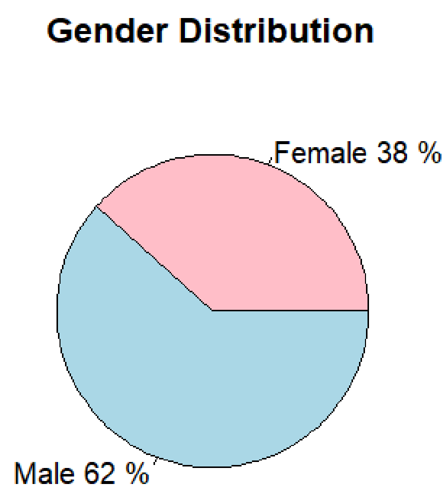
The focus of this topic is to examine the level of social media influence on physical activities among Universiti Teknologi Malaysia (UTM) students. Through the data collections from at least 60 respondents,our expectation leads to negative effects caused by spending too much time on social media to the point of neglecting physical activitiesThis report will discuss the analysis of data using R programming language to observe the existence of trend within the population studied

2. Data Collection

In the phase of data collection, the questionnaire was conducted through a Google form. x respondent feedback has been successfully collected, consisting of students from Universiti Teknologi Malaysia (UTM).The questionnaire used in this study consists of both categorical and quantitative variables.The categorical variables which are the gender,preferred social media platform and faculty,year of study ,time of engagement in physical activity and primary type of physical activity,while the quantitative variable includes the number of days per week engage in physical activity,hours per week spend on physical activities and hours per day spend on social media.

3. Data Analysis

A) Categorical Data

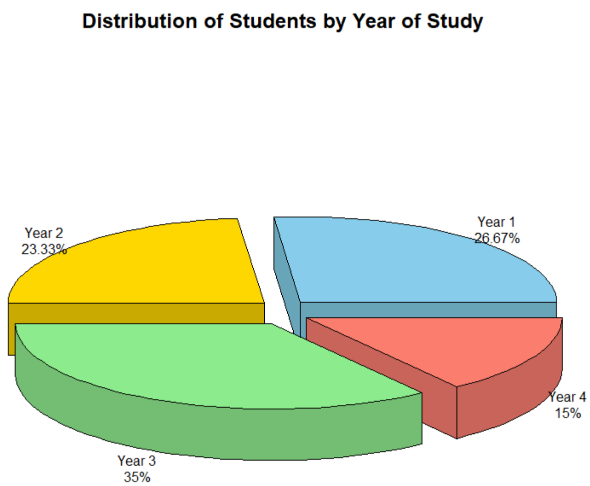


Variable: Gender

- **Sample Size :** 60 respondents
- **Proportion:**

Category	Frequency	Percentage (%)
Male	37	61.67
Female	23	38.33

Interpretation: Out of the 60 total students surveyed, the majority of respondents were Male, making up 61.67% of the sample, while the remaining 38.33% were Female

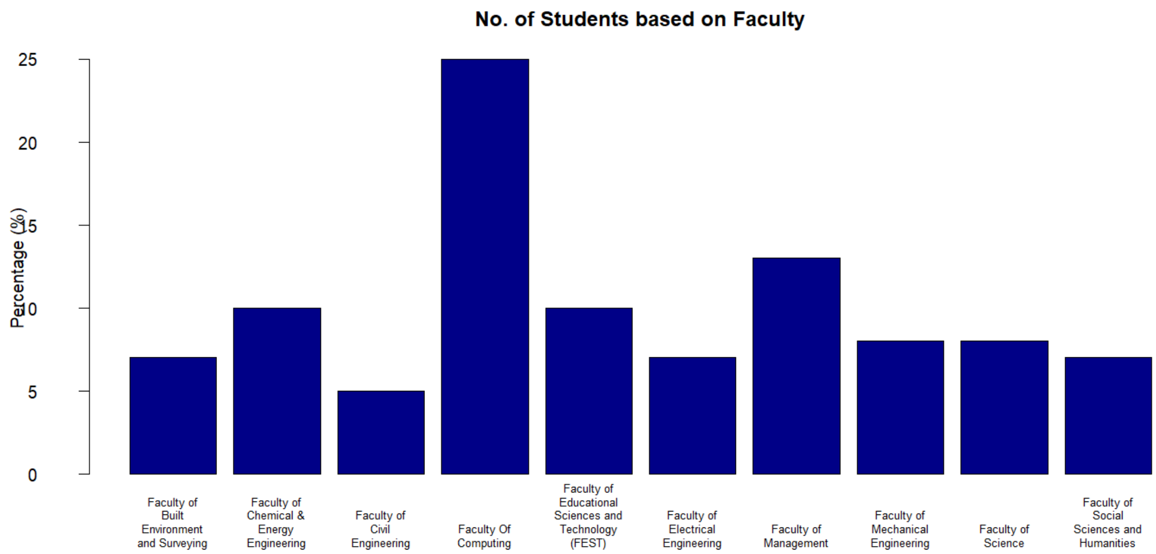


Variable: Year of Study

- **Sample Size :** 60 respondents
- **Proportions:**

Category	Frequency	Percentage (%)
Year 1	16	26.67
Year 2	14	23.33
Year 3	21	35.00
Year 4	9	15.00

Interpretation: The pie chart illustrates the distribution of our respondents across different academic years. The largest group of participants comes from Year 3, making up 35.00% of the total sample (21 students). Year 1 and Year 2 follow behind with 26.67% and 23.33% respectively. Conversely, Year 4 students represent the smallest portion of our survey at just 15.00%, indicating that our data captures more input from lower and mid-degree students.



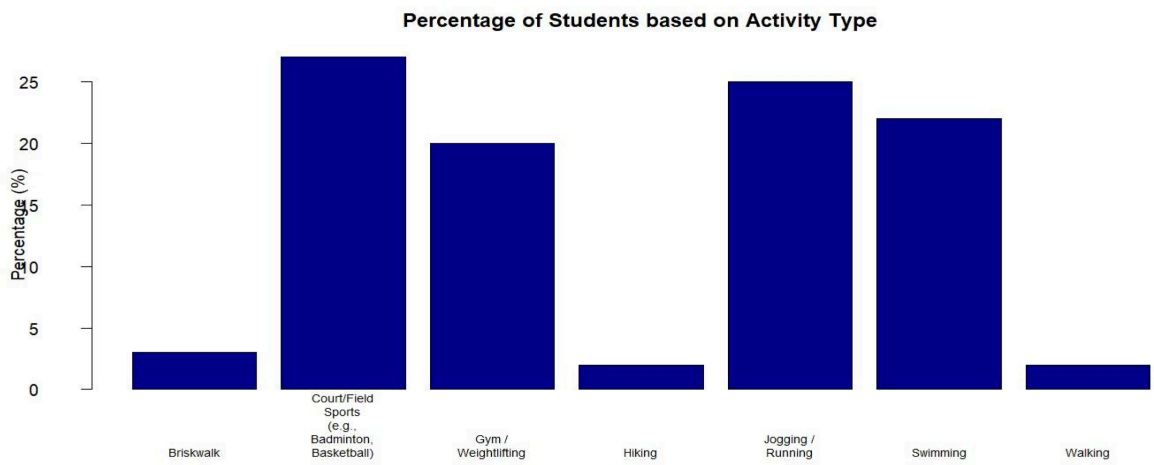
Variable: Faculty

- **Sample Size:** 60 respondents
- **Proportions (Frequency):**

Category	Frequency	Percentage (%)
Faculty of Computing	15	25.0
Faculty of Management	8	13.0
Faculty of Chemical & energy Engineering	6	10.0
Faculty of Educational Sciences and Technology (FEST)	6	10.0
Faculty of Mechanical Engineering	5	8.0
Faculty of Science	5	8.0
Faculty of Built Environment and Surveying	4	7.0
Faculty of Electrical Engineering	4	7.0
Faculty of Social Sciences	4	7.0

and Humanities		
Faculty of Civil Engineering	3	5.0

Interpretation: Out of the 60 students surveyed, our respondents come from a diverse range of academic backgrounds. The highest proportion of students belongs to the Faculty of Computing, representing 25% of our sample (15 students). The second largest group is from the Faculty of Management, at 13% (8 students). The remaining students are distributed across the other 8 faculties as listed above, showing a wide spread of participation across the university.

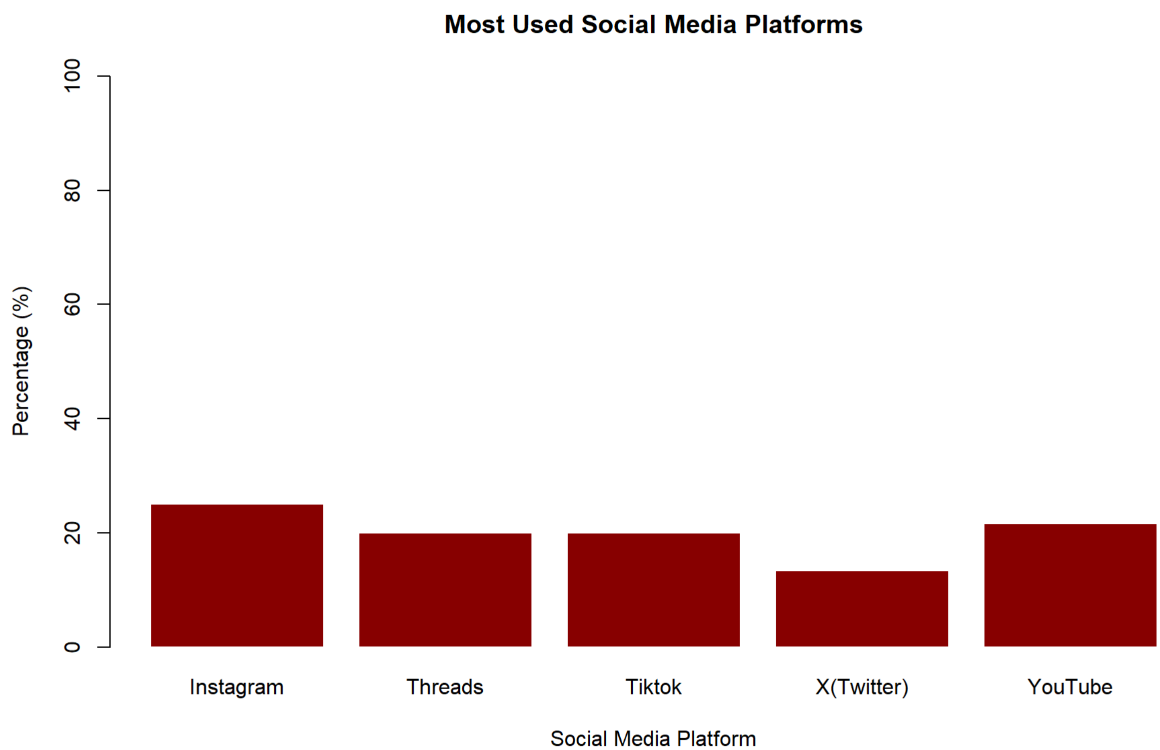


Variable: Type Of Activity

- **Sample SIze =** 60 Respondents
- **Proportions:**

Category	Frequency	Percentage(%)
Briskwalk	2	3.33
Field Sports	16	26.67
Gym	12	20.00
Hiking	1	1.67
Jogging	15	25.00
Swimming	13	21.67
Walking	1	1.67

- **Interpretation :** Based on data we had collected from 60 respondents UTM students, Field Sports is the most popular type of activity with the highest percentage of 26.67%. Jogging and Swimming activities also recorded high percentages, 25.00% and 21.67% of the total sample. On the other hand, activities such as Hiking and Walking are the least with a frequency of only one respondent for each category (1.67%). Overall, the majority of students prefer to engage in team sports or high-intensity activities compared to ordinary activities such as walking.

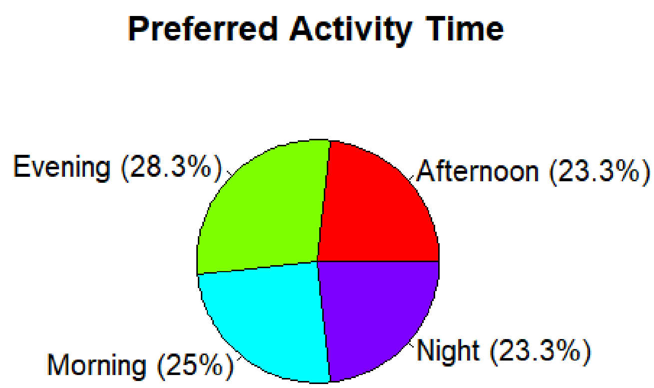


Variable: Social Media Platform

- **Sample Size :** 60 respondents
- **Proportion:**

Category	Frequency	Percentage(%)
Instagram	15	25.0
Youtube	13	21.7
Threads	12	20.0
Tiktok	12	20.0
X(Twitter)	8	13.3

- **Interpretation :** This data show that Instagram is the most utilized platform, accounting for 25% of the respondents' time, followed closely by YouTube at 21.7%. Meanwhile, Threads and TikTok show equal engagement levels at 20% each, leaving X (Twitter) as the least used platform among the group at 13.3%.

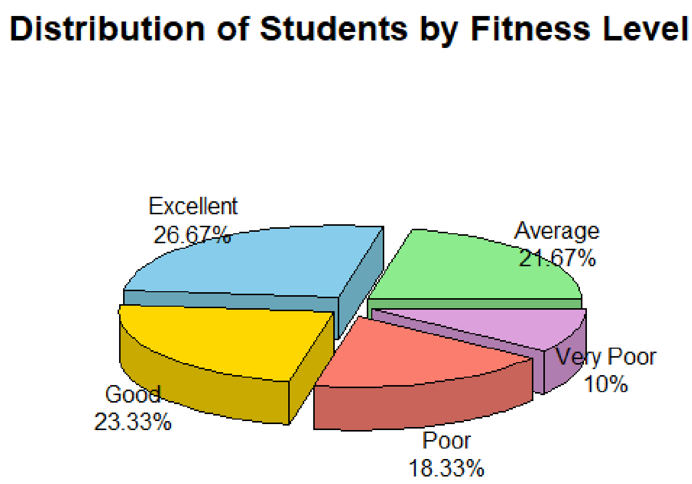


Variable: Activity Time

- **Sample Size =** 60 Respondents
- **Proportions :**

Category	Frequency	Percentage(%)
Morning	17	25.00
Evening	15	28.33
Afternoon	14	23.33
Night	14	23.33

- **Interpretation :** Based on the results we had collected from 60 respondents UTM students, the evening period has the highest percentage which is 28.33%. The morning period is the second highest choice which is 25.00%. Next, the afternoon and night periods which share the same value, which is 23.33%. Overall, this data represents that the distribution of respondents' choices is almost balanced, with a slightly higher percentage in the evening.

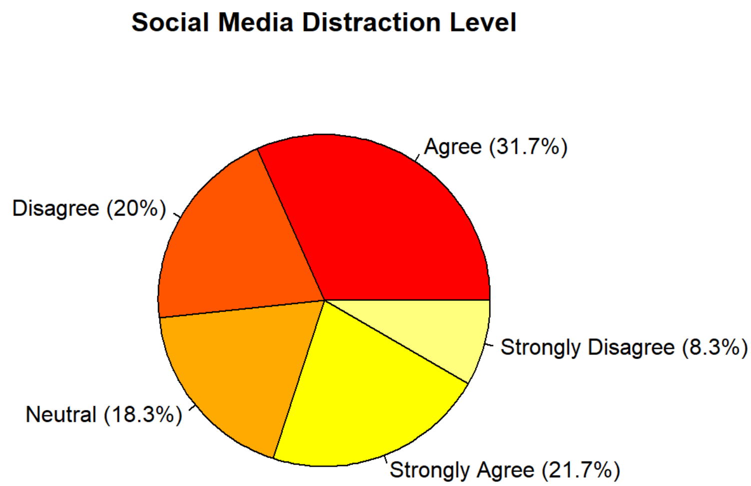


Variable: Fitness Level

- **Sample Size : 60 respondents**
- **Proportions:**

Category	**Frequency**	**Percentage (%)**
Excellent	16	26.67
Good	14	23.33
Average	13	21.67
Poor	11	18.33
Very Poor	6	10.0

Interpretation: The 3D pie chart illustrates the self-reported fitness levels of the participants. The largest group of respondents considers their fitness level to be Excellent, making up 26.67% of the total sample, closely followed by Good at 23.33%. Conversely, only 10.00% of students rated their fitness as Very Poor, indicating that the majority of our surveyed population views their physical health relatively positively.



Variable: Social Media Distract

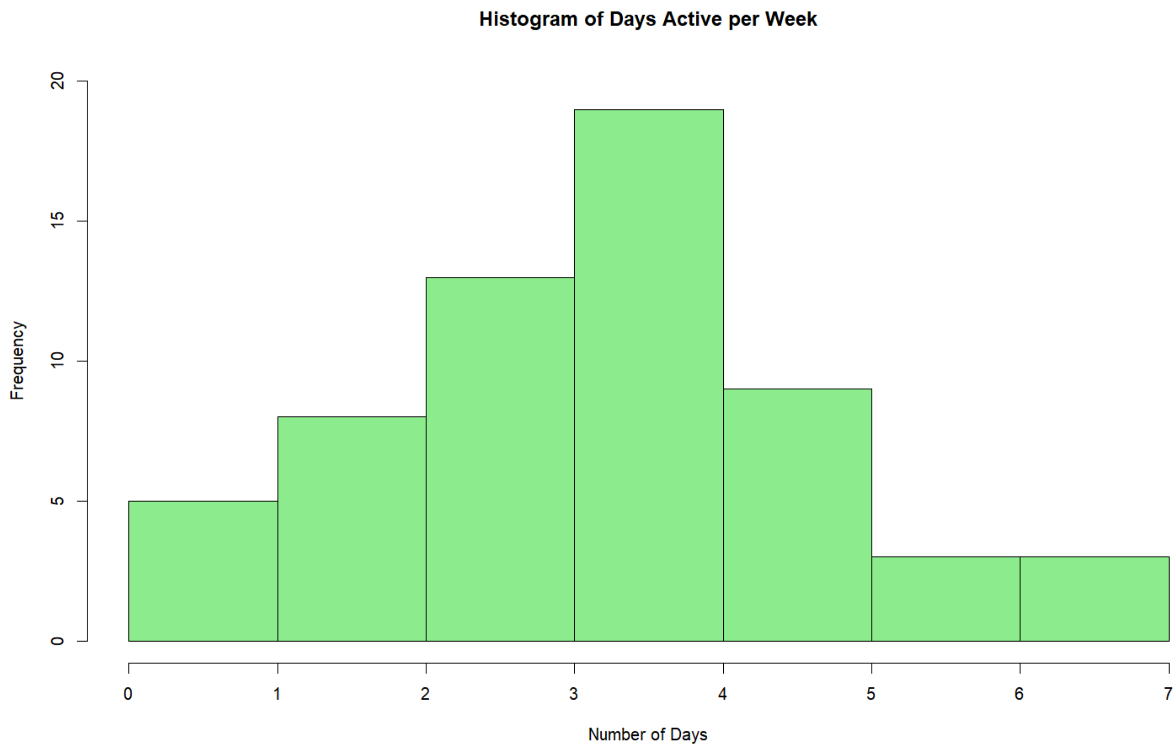
- **Sample Size :** 60 respondents
- **Proportion:**

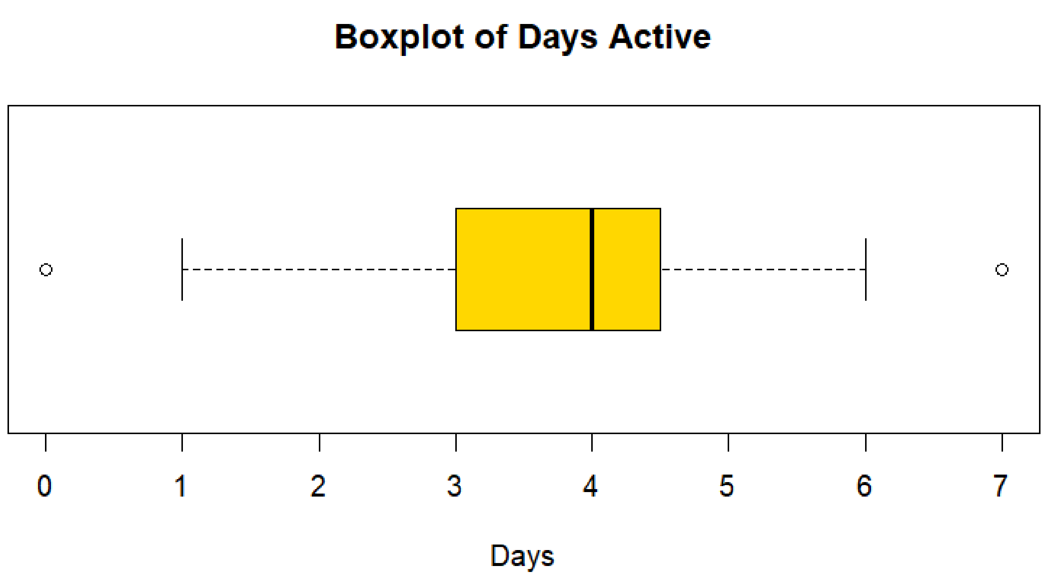
Category	Frequency	Percentage (%)
Strongly Disagree	13	21.7
Disagree	12	20.0
Neutral	11	18.3
Agree	19	31.7
Strongly Agree	5	8.3

Interpretation: The data suggests a divided perspective on social media distraction, with the largest portion of respondents (31.7%) agreeing that it is a disruptive factor. However, a significant combined minority of 41.7% expressed some level of disagreement, indicating that the experience of distraction is not universal across the surveyed group.

.

B) Numerical Data



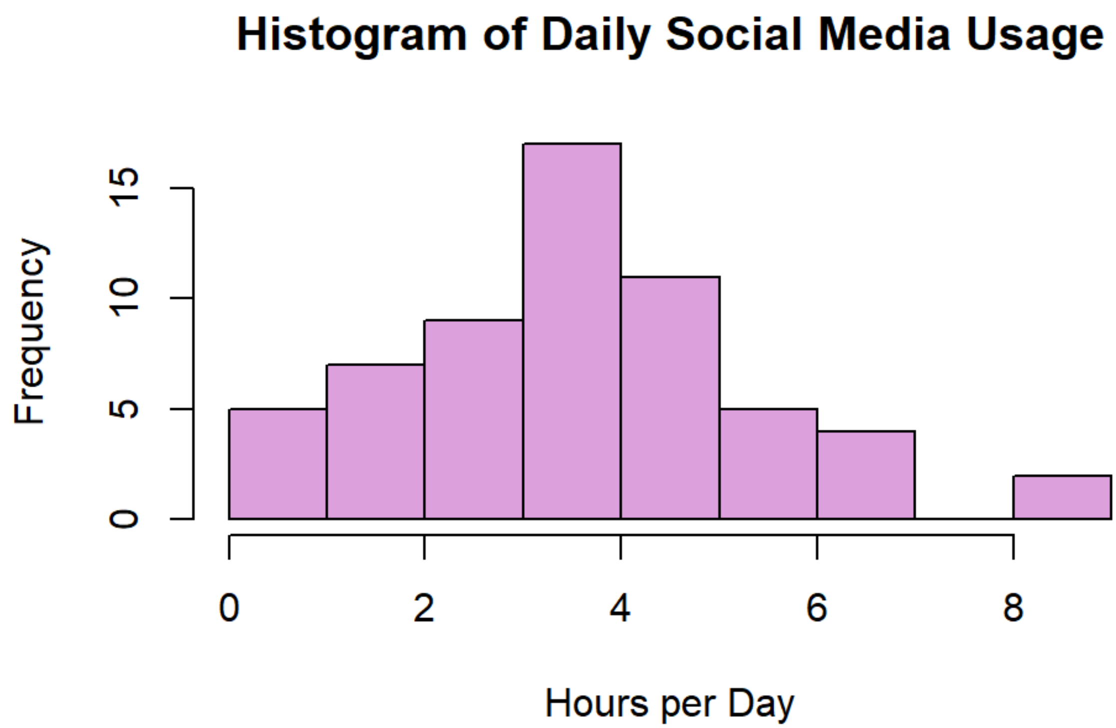


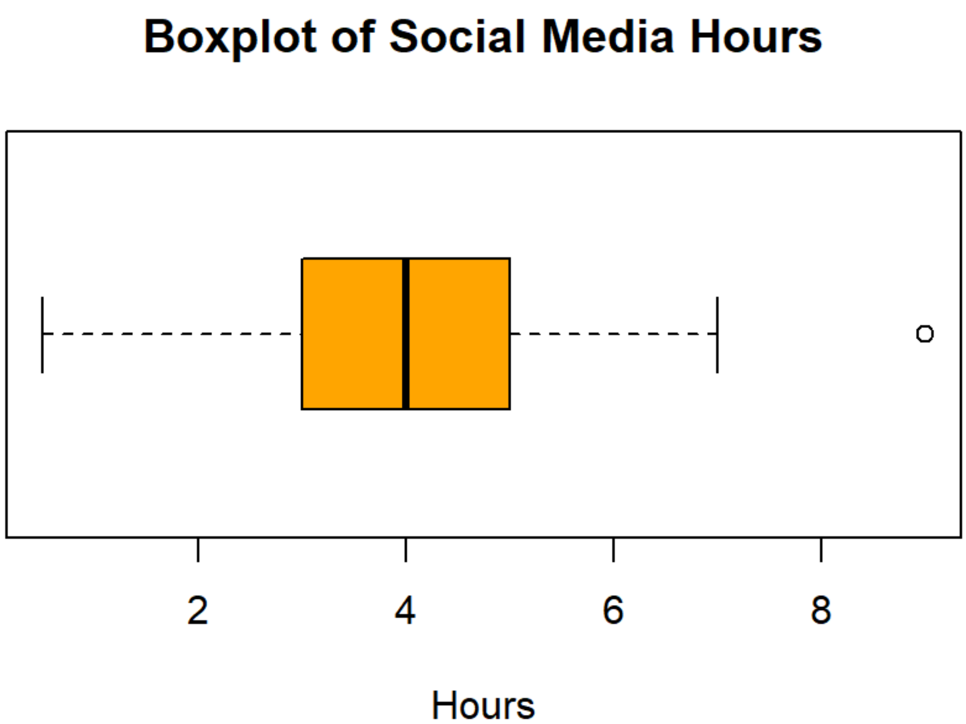
Variables : Days of active

- **Sample Size = 60 respondents**
- **Measure of Central Tendency:**

Statistic	Value
Mean	3.73
Median	4.00
Standard Deviation	1.56
Minimum	0.00
Maximum	6.00

Interpretation: Based on the data of Histogram above, most respondents are active in sports for 3 to 4 days in a week as shown. The mean value is 3.73 days, while the median value is 4.00 days per week. The boxplot chart shows that there are outliers or extreme values at 0 and 7, it means that there are only a few students who exercise very rarely or very frequently. Overall, the majority of students have a fitness level consistent with an average of almost 4 active days per week.


Histogram of Daily Social Media Usage
Frequency
0
5
10
15
0
2
4
6
8
Hours per Day

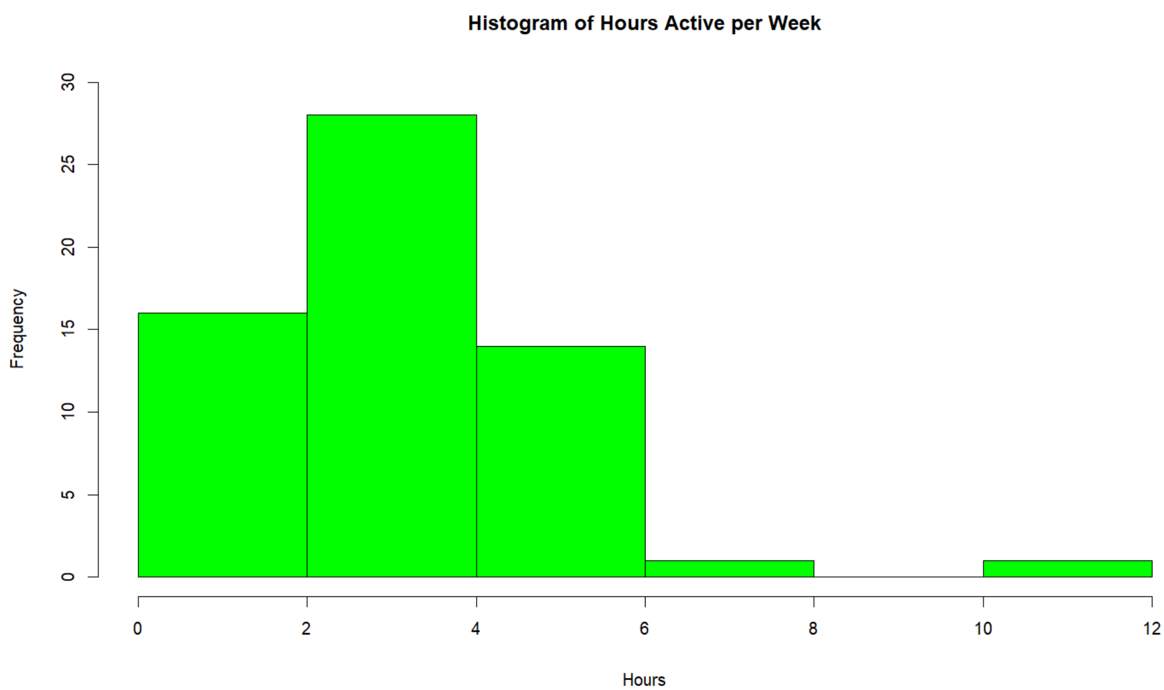


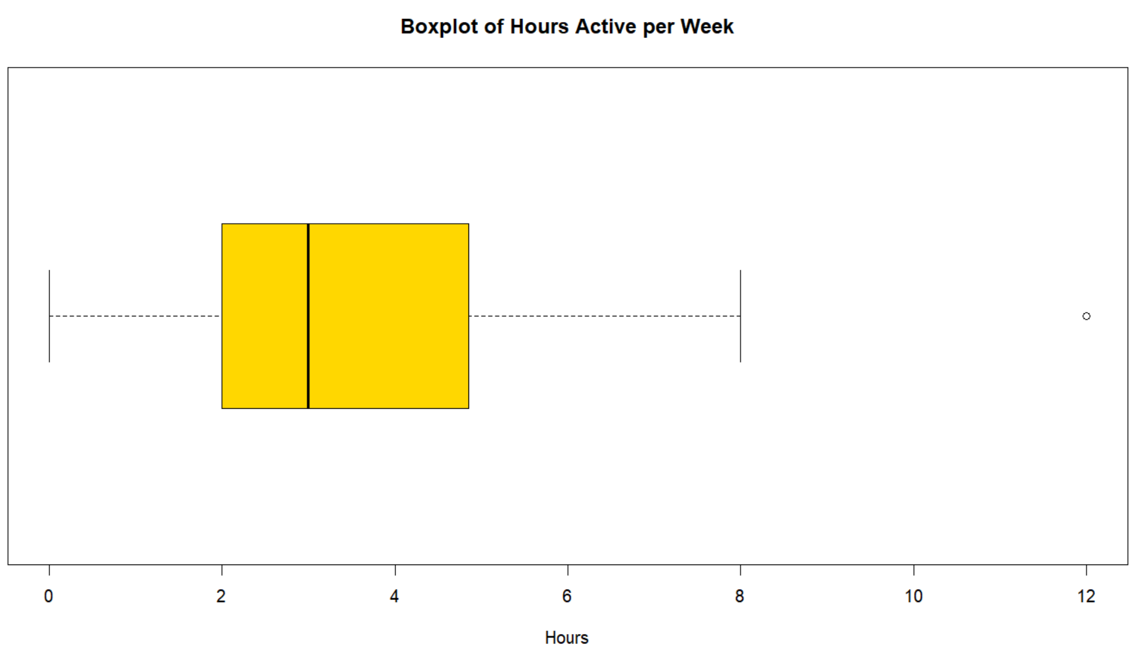
Variable: Hours Active

- **Sample Size :** 60 respondents
- **Proportion:**

Statistic	Value
Mean	4.02
Median	4.00
Standard Deviation	1.85
Minimum	0.5
Maximum	7.00

Interpretation: The histogram reveals a distribution that peaks between 3 and 4 hours, indicating that while most usage clusters within a 2 to 5-hour range, the data is positively skewed (right-skewed). This tail toward the 8 to 9-hour mark suggests that a small group of high-frequency users is pulling the statistical average upward, despite the majority following a more moderate, roughly normal pattern.


Histogram of Hours Active per Week
Frequency
0
5
10
15
20
25
30
0
2
4
6
8
10
12
Hours


Boxplot of Hours Active per Week
0
2
4
6
8
10
12
Hours

Statistic	Value
Mean	3.60
Median	3
Standard Deviation	1.94
Minimum	0.00
Maximum	8.00

Outliers: One respondent is considered an outlier, reporting a significantly high activity level of 12.00 hours per week.

Based on the survey data from 60 respondents (n = 60), the variable "Hours Active per Week" provides significant insight into the fitness habits of the student population. By comparing the visual data from the Histogram and Boxplot alongside the calculated statistics, we observe a clear right-skewed distribution. This skewness is mathematically evidenced by the Mean (3.60) being higher than the Median (3.00), as the average is being pulled upward by high-value data points.

The histogram shows that the highest frequency of students (28 individuals) exercise for 2 to 4 hours per week, which aligns with the central "box" of the boxplot. While the Standard Deviation of 1.94 suggests a relatively moderate spread, the boxplot highlights a notable gap between the Maximum of 8.00 hours and a single outlier at 12.00 hours. This outlier represents a respondent whose activity level is four times the median, likely a student athlete or fitness enthusiast. Consequently, the median of 3.00 hours serves as a more robust measure of the "typical" student's activity level, as it remains unaffected by the extreme 12-hour value. Overall, the data illustrates that while the majority of students maintain a consistent, moderate exercise routine, very few extend their physical activity beyond 8 hours a week.

4. Conclusion

Based on the responses from 60 university students from UTM, the study shows that social media is widely used among students, with an average usage of about 4 hours per day. However, most students still participate in physical activities several times per week and spend around 3 to 4 hours exercising. This suggests that although social media is an important part of students' daily lives, many students still find their perfect time to maintain an active lifestyle.

From the google form , we also found that social media can affect physical activity habits. Most of the respondents agree that social media can distract them from exercising. Based on this, we can conclude that excessive use of social media may reduce the time for physical activity, which can negatively affect students' fitness and health. Therefore, as students, we need to know how to balance our social media use with regular exercise to maintain a healthy lifestyle while enjoy.